

COLLABORATIVE AMR FOR WAREHOUSE-TO-AISLE STOCK REPLENISHMENT

WEEK 4 REPORT - GROUP 7

PROGRESS SO FAR

Establish bidirectional **communication between ROS 1 and Unity** physics simulation, publish real-time **odometry/TF**, and achieve closed-loop **autonomous navigation from Point A to Point B**.

Key Achievements:

- Implemented Unity planar odometry publisher (/odom, /tf, /clock)
- Developed and verified the go_to_point closed-loop controller node in ROS.
- Validated trajectory tracking and precision stopping in RViz and Unity.

SYSTEM ARCHITECTURE & DATA FLOW

Platform Stack:

Simulation: Unity (Windows)/ Unity Ubuntu - AMR 3D model, physics, warehouse environment

Middleware: ROS–TCP Connector & ros_tcp_endpoint.

Core Stack: ROS 1 Noetic running inside a Docker container (WSL2).

Communication Pipelines:

- **Unity → ROS:**

- **/clock:** Simulation time synchronization (ClockMsg)
 - **rosparam set /use_sim_time true:** ROS nodes completely ignore the system clock and instead listen exclusively to the /clock topic publishing from Unity
- **/odom:** Ground-truth planar pose and twist (OdometryMsg)
- **/tf:** Dynamic transform odom -> base_footprint.

- **ROS → Unity:**

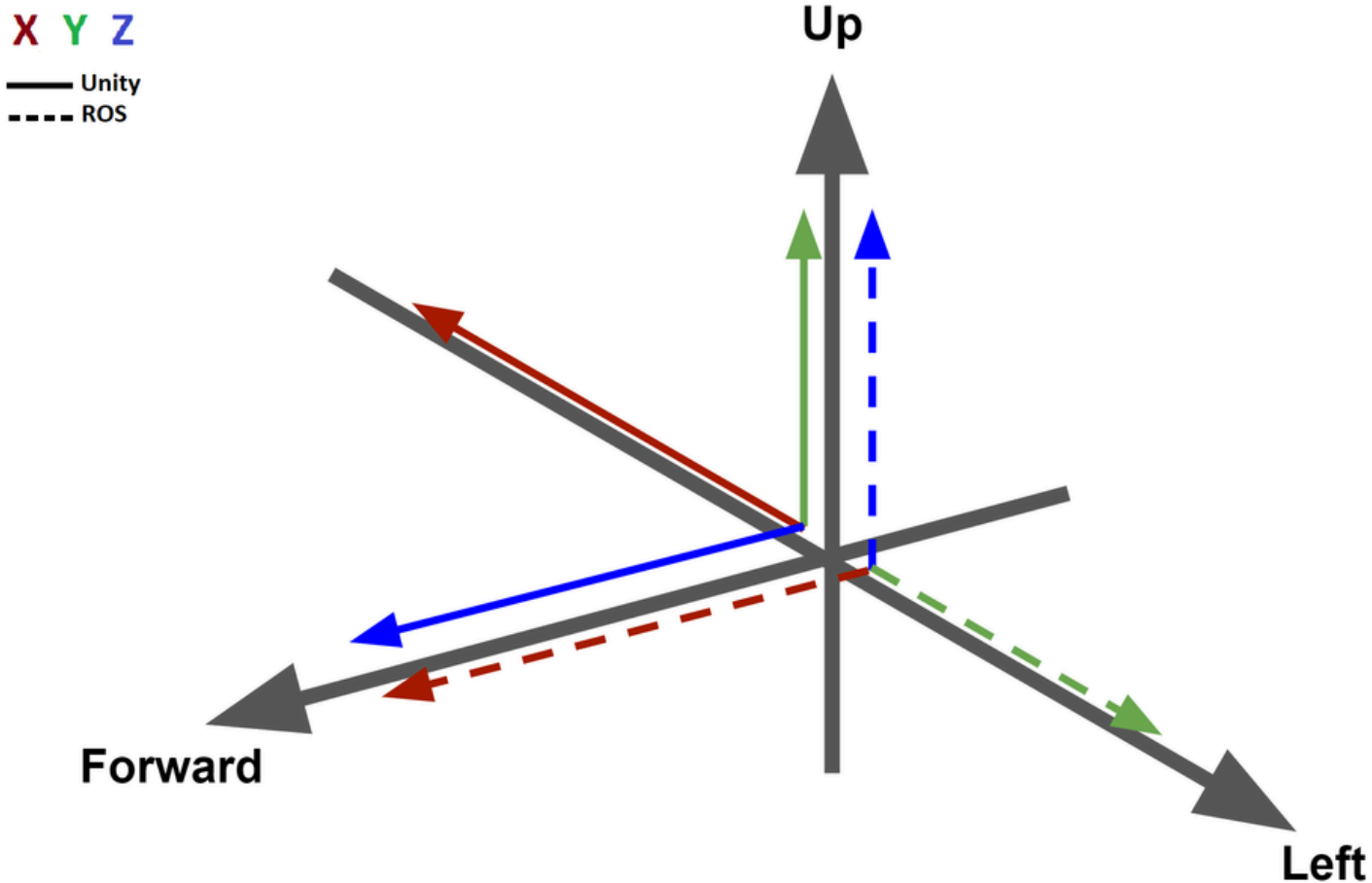
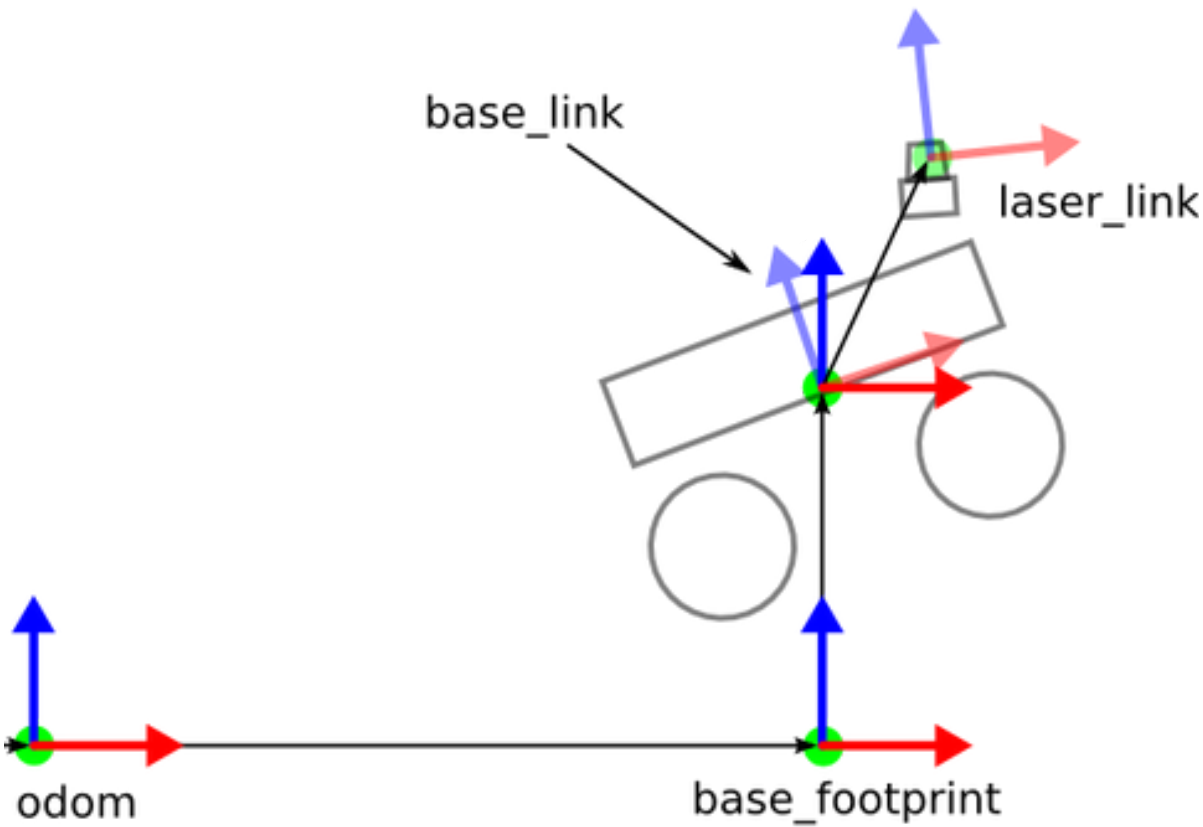
- **/cmd_vel:** Velocity demands (Twist – linear.x, angular.z).

```
root@705eb24e72f3:/catkin_ws# rostopic list
/clicked_point
/clock
/cmd_vel
/initialpose
/lift_cmd
/move_base_simple/goal
/odom
/rosout
/rosout_agg
/scan
/tf
/tf_static
```

COORDINATE FRAMES & TF ALIGNMENT

Frame Tree Setup:

- Anchor frame: **odom** (instantiated at robot spawn point).
- Base frame: **base_footprint** derived from moving **base_link** ArticulationBody.



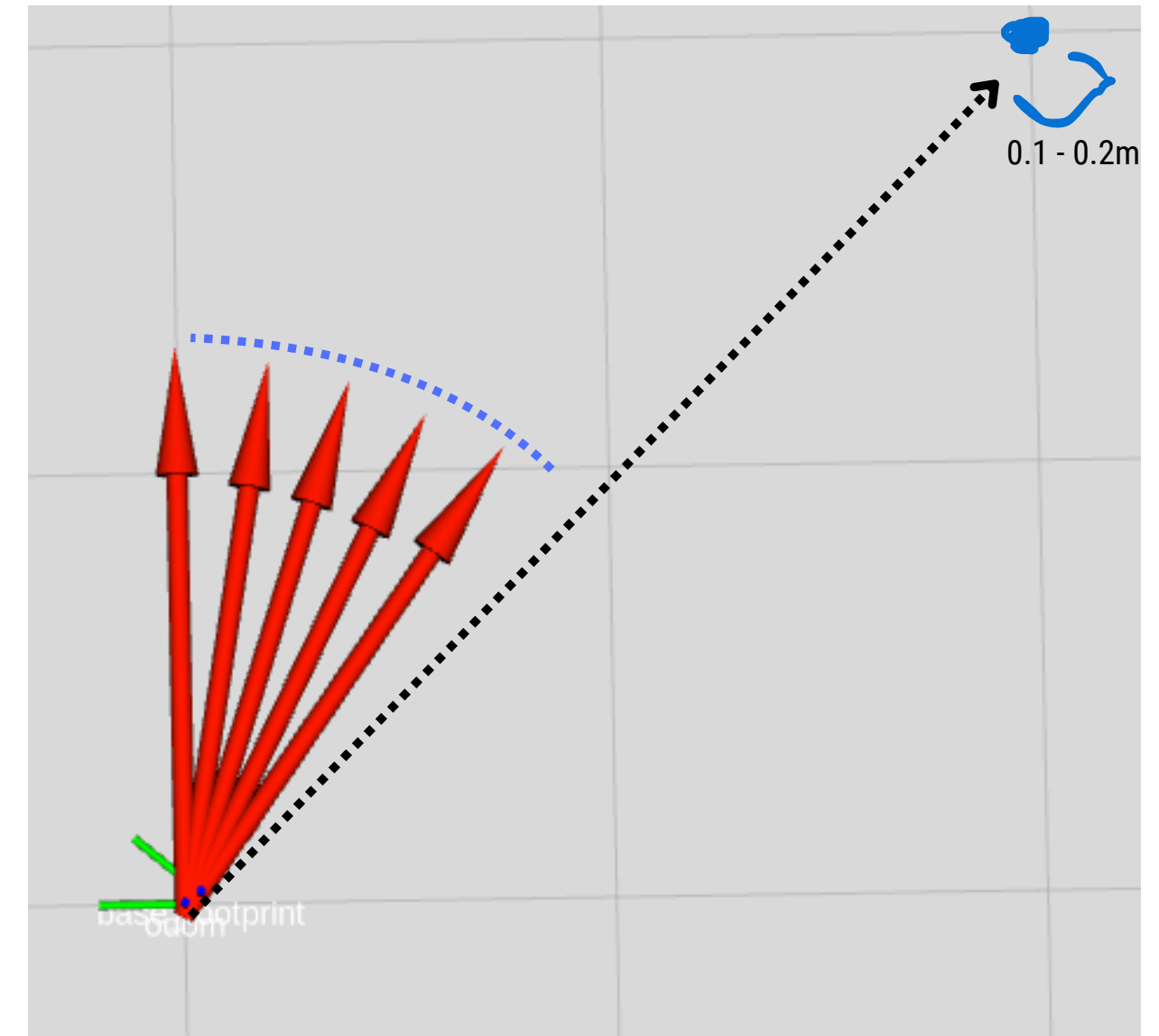
	ROS	UNITY
COORDINATE SYSTEM	FLU(Forward-Left-Up)	RUF(Right-Up-Forward)
POSITIVE ROTATION	Counter-Clockwise (CCW)	Clockwise (CW)

AUTONOMOUS NODE IMPLEMENTATION (GO_TO_POINT)

Control Paradigm: Closed-loop proportional tracking based on dynamic /odom feedback.

Navigation Logic:

1. **Orientation alignment:** If heading error $|\Delta\theta| > 0.35 \text{ rad}$, rotate in place.
2. **Drive & throttle:** When aligned, drive forward $v_{\max} = 0.2 \text{ m/s}$, scaling speed inversely with distance.
3. **Safety watch:** Halts vehicle if odometry gap or timestamp age exceeds 0.5s.
4. **Goal acceptance:** Triggers stop sequence when within tolerance radius 0.1 - 0.2m



EXPERIMENTAL RESULTS & VERIFICATION

Test Case: Moving to goal X = 2.0m, Y = -2.0m with tolerance 0.10m.

Observed Execution:

- **ROS Nodes:** Active bridge between */unity_endpoint*, */go_to_point*, and */rviz*.
- **Velocity:** Smooth initial rotation followed by linear translation on */cmd_vel*.
- **Path Verification:** RViz trajectory arrows and Unity viewport showed identical motion and final stopping stance within tolerance.

```
root@705eb24e72f3:/catkin_ws# roslaunch amr_navigation go_to_point.launch goal_x:=2.0 goal_y:=-2.0
SUMMARY
=====

PARAMETERS
* /go_to_point/goal_x: 2.0
* /go_to_point/goal_y: -2.0
* /go_to_point/max_angular: 0.5
* /go_to_point/max_linear: 0.2
* /go_to_point/tolerance: 0.1
* /roscdistro: noetic
* /rosversion: 1.17.4

NODES
/
  go_to_point (amr_navigation/go_to_point.py)

ROS_MASTER_URI=http://127.0.0.1:11311

process[go_to_point-1]: started with pid [8411]
[INFO] [1790516533.349355, 0.000000]: Goal in odom: (2.00, -2.00), tolerance 0.10 m
[WARN] [1790516533.350820, 0.000000]: Waiting for /odom
█
```

default.rviz* - RViz (on 705eb24e72f3)

File Panels Help

Interact Move Camera Select Focus Camera Measure 2D Pose Estimate 2D Nav Goal Publish Point

Displays

- Global Options
 - Fixed Frame: odom
 - Background Color: 48; 48; 48
 - Frame Rate: 30
 - Default Light: ☒
- Global Status: Ok
- Grid
 - Status: Ok
 - Reference Frame: <Fixed Frame>
 - Plane Cell Count: 10
 - Normal Cell Count: 0
 - Cell Size: 1
 - Line Style: Lines
 - Color: 160; 160; 164
 - Alpha: 0.5
 - Plane: XY
 - Offset: 0; 0; 0
- TF
 - Status: Ok
 - Show Names: ☒
 - Show Axes: ☒
 - Show Arrows: ☒
 - Marker Scale: 1
 - Marker Alpha: 1
 - Update Interval: 0
 - Frame Timeout: 15
 - Filter (whitelist):
 - Filter (blacklist):
- Frames
 - All Enabled: ☐
 - base_footprint: ☒
 - odom: ☒
- Tree
- Odometry
 - Status: Ok
 - Topic: /odom
 - Unreliable: ☐
 - Queue Size: 10
 - Position Tolerance: 0.1
 - Angle Tolerance: 0.1
 - Keep: 1
 - Shape: Arrow
 - Covariance: ☒

Add Duplicate Remove Rename

Time

Pause Synchronization: Off ROS Time: 1.02 ROS Elapsed: 0.67 Wall Time: 7009.98 Wall Elapsed: 68.10

Unity - amr_ware_house - SampleScene - Windows, Mac, Linux - Unity 2022

Assets GameObject Component Services Robotics Help Window

UnityVCS

Scene

Center Local

Game

Display 1 Free Aspect

Console

Assets > Scenes

SampleSc...

default.rviz* - RViz (on 705eb24e72f3)

File Panels Help

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 - Line Style: Lines
 - Color: 160; 160; 164
 - Alpha: 0.5
 - Plane: XY
 - Offset: 0; 0; 0
- TF
 - Status: Ok
 - Show Names: ☒
 - Show Axes: ☒
 - Show Arrows: ☒
 - Marker Scale: 1
 - Marker Alpha: 1
 - Update Interval: 0
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 - All Enabled: ☐
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- Odometry
 - Status: Ok
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 - Queue Size: 10
 - Position Toleran...: 0.1
 - Angle Tolerance: 0.1
 - Keep: 1
 - Shape: Arrow
 - Covariance: ☒

Add Duplicate Remove Rename

Time

Pause Synchronization: Off ROS Time: 230.13 ROS Elapsed: 229.82 Wall Time: 6871.43 Wall Elapsed: 238.12

Reset Left-Click: Rotate. Middle-Click: Move X/Y. Right-Click/Mouse Wheel:: Zoom. Shift: More options. 31 fps

Unity - amr_ware_house - SampleScene - Windows, Mac, Linux - Unity 2022

Assets GameObject Component Services Robotics Help Window

UnityVCS

Scene

Center Local

Game

Display 1 Free Aspect

ROS IP: 127.0.0.1:10000

Console

Assets > Scenes

SampleSc...

roller] Initialized! Friction tuned (casters=0, wheels=1), Keyboard (WASD + R/F) active.

***ROS send movement instructions of type
geometry_msgs/Twist message to Unity for simulation***

```
root@705eb24e72f3:/catkin_ws# rostopic echo /cmd_vel
linear:
  x: 0.2
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: -0.009279019867284688
---
linear:
  x: 0.2
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: -0.007308254305974926
---
linear:
  x: 0.2
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: -0.007308254305974926
---
linear:
  x: 0.2
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: -0.007308254305974926
---
```

***tf_echo odom base_footprint matching
real-time vehicle position and yaw***

```
root@705eb24e72f3:/catkin_ws# rosrun tf tf_echo odom base_footprint
At time 1.751
- Translation: [-0.023, 0.002, 0.000]
- Rotation: in Quaternion [0.000, 0.000, -0.137, 0.991]
              in RPY (radian) [0.000, 0.000, -0.276]
              in RPY (degree) [0.000, 0.000, -15.798]
At time 1.751
- Translation: [-0.023, 0.002, 0.000]
- Rotation: in Quaternion [0.000, 0.000, -0.137, 0.991]
              in RPY (radian) [0.000, 0.000, -0.276]
              in RPY (degree) [0.000, 0.000, -15.798]
At time 2.202
- Translation: [-0.028, 0.004, 0.000]
- Rotation: in Quaternion [0.000, 0.000, -0.202, 0.979]
              in RPY (radian) [0.000, 0.000, -0.407]
              in RPY (degree) [0.000, 0.000, -23.324]
At time 2.879
- Translation: [-0.014, -0.005, 0.000]
- Rotation: in Quaternion [0.000, 0.000, -0.288, 0.958]
              in RPY (radian) [0.000, 0.000, -0.584]
              in RPY (degree) [0.000, 0.000, -33.467]
At time 4.041
- Translation: [0.114, -0.116, 0.000]
- Rotation: in Quaternion [0.000, 0.000, -0.380, 0.925]
              in RPY (radian) [0.000, 0.000, -0.780]
              in RPY (degree) [0.000, 0.000, -44.713]
At time 5.168
- Translation: [0.231, -0.233, 0.000]
- Rotation: in Quaternion [0.000, 0.000, -0.382, 0.924]
              in RPY (radian) [0.000, 0.000, -0.783]
              in RPY (degree) [0.000, 0.000, -44.885]
At time 6.071
- Translation: [0.325, -0.326, 0.000]
- Rotation: in Quaternion [0.000, 0.000, -0.378, 0.926]
              in RPY (radian) [0.000, 0.000, -0.776]
              in RPY (degree) [0.000, 0.000, -44.445]
At time 6.998
- Translation: [0.421, -0.420, 0.000]
- Rotation: in Quaternion [0.000, 0.000, -0.376, 0.927]
              in RPY (radian) [0.000, 0.000, -0.771]
              in RPY (degree) [0.000, 0.000, -44.182]
```

PLAN FOR UPCOMING WEEK

- **Verify requirements and deliverables of Milestone 1** to see what is still missing and needs improvement → **finalize Milestone 1.**
- **Sensor integration:**
 - simulated LiDAR pipeline via /scan with proper robot self-collision filtering
 - visualize LaserScan successfully on RViz
- **Build scene:** Build warehouse scene/environment for testing